MATH 128A - HW 2

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1(a). Compute the absolute error and relative error in approximations of p by p^*.

- $\pi \approx 3.141592$,
- $\frac{22}{7} \approx 3.142857$.

$$E_{\text{abs}} = |p - p^*| = \left|\pi - \frac{22}{7}\right| \approx |3.141592 - 3.142857| = 0.001264$$

$$E_{\text{rel}} = \frac{|p - p^*|}{|p|} \approx \frac{0.001264}{3.141592} \approx 4.025 \times 10^{-4}$$

2(d). Compute the absolute error and relative error in approximations of p by p^*.

- $p = 9! = 362880$
- $p^* = \sqrt{18\pi}\left(\frac{9}{e}\right)^9 \approx \sqrt{56.5487} \times (3.3109)^9 \approx 7.5199 \times 473811.4867 \approx 359536.8728$

$$E_{\text{abs}} = |p - p^*| = |362880 - 359536.8728| \approx 3343.1272$$

$$E_{\text{rel}} = \frac{|p - p^*|}{|p|} = \frac{3343.1272}{362880} \approx 9.213 \times 10^{-3}$$

3(b). Suppose p^* must approximate p with relative error at most 10^{-3}. Find the largest interval in which p^* must lie for each value of p.

$$\frac{|p^* - p|}{|p|} \leq 10^{-3} \quad \Longrightarrow \quad |p^* - p| \leq 10^{-3}|p|.$$

So p^* must lie in $[p(1 - 10^{-3}),\, p(1 + 10^{-3})]$.

For $p = 900$:

$$10^{-3} \cdot 900 = 0.9,$$

so

$$p^* \in [900 - 0.9,\ 900 + 0.9] = [899.1,\ 900.9].$$

6(a). Use three-digit rounding arithmetic to perform the following calculations. Compute the absolute error and relative error with the exact value determined to at least five digits.

- 133 is already a three-digit number.
- 0.921 is also a three-digit number.

Exact sum:

$$p = 133 + 0.921 = 133.921$$

Rounding 133.921 to three significant digits: $p^* = 133.921 \approx 134$

- **Absolute error**:

$$E_{\text{abs}} = |p - p^*| = |134 - 133.921| = 0.079.$$

- **Relative error**:

$$E_{\text{rel}} = \frac{E_{\text{abs}}}{|p|} = \frac{0.079}{133.921} \approx 5.9 \times 10^{-4}.$$

12. The number e can be defined by $e = \sum_{n=0}^{\infty} \frac{1}{n!}$, where $n! = n(n-1)\cdots 2 \cdot 1$ for $n \neq 0$ and $0! = 1$. Compute the absolute error and relative error in the following approximations of e:

$$e = \sum_{n=0}^{\infty} \frac{1}{n!} \approx 2.718281828$$

(a) $\sum_{n=0}^{5} \frac{1}{n!}$

- $n = 0$: $\frac{1}{0!} = 1$
- $n = 1$: $\frac{1}{1!} = 1$
- $n = 2$: $\frac{1}{2!} = \frac{1}{2} = 0.5$
- $n = 3$: $\frac{1}{3!} = \frac{1}{6} \approx 0.1666666667$
- $n = 4$: $\frac{1}{4!} = \frac{1}{24} \approx 0.0416666667$
- $n = 5$: $\frac{1}{5!} = \frac{1}{120} \approx 0.0083333333$

Add the terms:

$$p^* = 1 + 1 + 0.5 + 0.1666666667 + 0.0416666667 + 0.0083333333 = 2.7166666667$$

- **Absolute error**:

$$E_{\text{abs}} = |e - p^*| \approx |2.718281828 - 2.716666667| \approx 0.001615161$$

- **Relative error**:

$$E_{\text{rel}} = \frac{E_{\text{abs}}}{|e|} \approx \frac{0.001615161}{2.718281828} \approx 5.94 \times 10^{-4}.$$

(b) $\sum_{n=0}^{10} \frac{1}{n!}$

- $n = 0$: 1
- $n = 1$: 1
- $n = 2$: 0.5
- $n = 3$: $\frac{1}{6} \approx 0.1666666667$
- $n = 4$: $\frac{1}{24} \approx 0.0416666667$
- $n = 5$: $\frac{1}{120} \approx 0.0083333333$
- $n = 6$: $\frac{1}{720} \approx 0.0013888889$
- $n = 7$: $\frac{1}{5040} \approx 0.0001984127$
- $n = 8$: $\frac{1}{40320} \approx 0.0000248016$
- $n = 9$: $\frac{1}{362880} \approx 0.0000027557$
- $n = 10$: $\frac{1}{3628800} \approx 0.0000002756$

Add the terms:

$$p^* \approx 2.718281801$$

- **Absolute error**:
$$E_{\text{abs}} = |2.718281828 - 2.718281801| \approx 2.7 \times 10^{-8}.$$
- **Relative error**:
$$E_{\text{rel}} = \frac{E_{\text{abs}}}{|e|} \approx \frac{2.7 \times 10^{-8}}{2.718281828} \approx 9.9327 \times 10^{-9}.$$

19. Use the 64-bit-long real format to find the decimal equivalent of the following floating-point machine numbers.

(a) `0 10000001010 1001001100000000000000000000000000000000000000000000`

- **Sign bit**: $s = 0 \rightarrow$ positive
- **Exponent**:
  - $10000001010_2 = 2^{10} + 2^3 + 2^1 = 1024 + 8 + 2 = 1034$
  - $e - 1023 = 1034 - 1023 = 11$
- **Mantissa**:
  - $f = 2^{-1} + 2^{-4} + 2^{-7} + 2^{-8} = 0.57421875$
  - $1 + f = 1.57421875$

Final value:

$$p = 1.57421875 \cdot 2^{11} = 1.57421875 \cdot 2048 = 3224$$

(b) `1 10000001010 1001001100000000000000000000000000000000000000000000`

- **Sign bit**: $s = 1 \rightarrow$ negative
- All other bits same as part (a)

So,

$$p = -3224$$

(c) `0 01111111111 0101001100000000000000000000000000000000000000000000000000`

- **Sign bit**: $s = 0 \rightarrow$ positive
- **Exponent**:
  - $01111111111_2 = 1023$
  - $e - 1023 = 0$
- **Mantissa**:
  - $f = 2^{-2} + 2^{-4} + 2^{-7} + 2^{-8} = 0.32421875$
  - $1 + f = 1 + 0.32421875 = 1.32421875$

Final value:

$$p = 1.32421875 \cdot 2^0 = 1.32421875$$

20(a). Find the next largest and smallest machine numbers in decimal form for the number given in Exercise 19a.

From Exercise (a),

$$x = +1.574218750 \times 2^{11} = 3224.0.$$

For IEEE-754 double precision the spacing at exponent e is $2^{\,e-52}$. Here $e = 11$, so

$$\text{ULP} = 2^{11-52} = 2^{-41}.$$

Thus the adjacent machine numbers are $x \pm \text{ULP}$.

Write them in the requested short form $d.ddddddddd \pm 2^{-dd}$:

$$\text{Next largest} = 3224.00000000 \; + \; 2^{-41} \qquad \text{and} \qquad \text{Next smallest} = 3224.00000000 \; - \; 2^{-41}$$